

mitter to the receiver, this system does not suffer from blocking in the LOS path.

Tracked system. In a tracked system, the receiver and the transmitter can track each other. Mechanical or electronic control needs to be implemented for effective tracking.

Terrestrial links. Terrestrial FSO system is used for building to building communication or metro area extension. This is primarily an LOS communication. It has applications in disaster recovery and x-haul of 3G/4G/5G mobile communications.

Space links. An inter-satellite link is a communication between two low Earth orbit (LEO) satellites. An inter-orbital link is established between LEO and geostationary Earth orbit (GEO) satellites. Tight alignment and tracking, low power consumption, and being lightweight are critical requirements for space applications.

The architecture of WOC system

A WOC system consists of three major subsystems: Transmitter, channel,

and receiver. The block diagram of an FSO communication system is shown in Fig. 3. The role of transmit optics and receive optics are analogous to the antenna used in RF communication.

Transmitter. The role of the transmitter is to convert the information to a signal form suitable for transmission over the channel. The modulator modulates the information signal and drives the laser driver. The laser diode acts as the optical source that converts the information in the electrical domain to an optical signal. The transmit optics collects the laser light and directs it towards the receiver.

Channel. For the WOC system, the channel can be vacuum, atmosphere, or water. In case of the atmospheric channel, the signal faces random attenuation due to effects like fog, haze, cloud, and rain. Atmospheric turbulence is caused by random fluctuations of the refractive index of the medium due to random temperature fluctuations. Atmospheric turbulence is the main limiting factor for the

performance of an FSO communication link.

Receiver. In the receiver, the incident optical signal is captured by the receive optics, and it focuses this signal on the photodetector. The photodetector converts the optical signal to its electrical form. Then the amplifier amplifies the signal and also increases the signal-to-noise ratio (SNR). Finally, the demodulator gives an estimate of the input signal.

Wavelengths used in WOC system

The choice of wavelength is dependent on multiple factors like atmospheric effects, attenuation, eye safety regulations, availability, cost of components, and background noise power. Overall wavelength bands are the same as used in optical fibre communication.

The majority of FSO communication systems are designed to operate in the windows of 780-850nm and 1520-1600nm. The benefit of using these bands is that the components are readily available as these are used

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RECTIFIER BRIDGES	Current	Voltage	Part Number	
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		1200 V – 2200 V	PSB 25T & PSD 25T PSB 25TN & PSD 25TN	
	36 A	200 V – 1000 V	PSB 36MB & PSD 36MT PSB 36MBN & PSD 36MTN	
		1200 V – 2200 V	PSB 36T & PSD 36T PSB 36TN & PSD 36TN	

Highlights:

- > Single phase and three phase uncontrolled bridges
- > Compact package, easy to mount with one screw
- > Connectors suitable for easy PCB mounting
- > Gold plated or nickel plated terminals
- > UL certified, RoHS and REACH conform and Halogen-Free

Nomenclature:

TN	Add ON T, MT, MB = Different chip categories, Gold Plated TN, MTN, MBN = Different chip categories, Nickel Plated	
25	Current Rating 25 = 25A	36 = 36A
PSD	Device Type PSD – Three Phase	PSB – Single Phase

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